

Improved cyclist and helmet detection using loss rank mining and YOLOv8

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Abstract— The growing emphasis organisations place on safety has resulted in several initiatives promoting bike helmet use. While interest in measuring bike helmet use has grown, measuring helmet usage is primarily a manual task done by human counters. While computer vision presents a promising solution to automating this task, the presence of vehicles resembling bicycles, such as mopeds, scooters and motorcycles can confuse models and cause inaccurate results. In addressing this, we propose using loss rank mining as a method of reducing false positive predictions, a mixed approach to training that ensures models learn both easy and difficult representations, and the use of trackers to count cyclists and helmets. Our approach has shown to reduce false positives by as much as half for bicycles, and to a much lesser extent for helmets. Additionally, MOTA increased by 8% when tracking bicycles with our approach, and by 16% when tracking helmets. Finally our counting results showed an accuracy of 90% for bicycles and a lower counting accuracy of 34% for helmets.

Index Terms—[Cyclist Safety, Object Detection, False Positive Suppression, Object Counting, Tracking, YOLOv8, Loss Rank Mining]

1 INTRODUCTION

Bicycle helmets are a primary safety measure for cyclists, and have been linked by several studies to be effective in reducing the risk of injury or fatality in the case of a crash [1, 2, 3, 4]. Owing to the benefits attributed to helmets, several initiatives have emerged that aim to quantify how often they are used. In particular, various municipalities and organisations are interested in understanding this statistic, which can help researchers uncover otherwise unknown information regarding cyclist safety. For example, understanding helmet use allows localities to determine the effectiveness of certain policies or campaigns [5, 6, 7, 8, 9]. Helmet usage may also be used as a local indicator of cyclist safety levels by relating the usage of helmets to severe injuries and fatalities in bicycle crashes [10, 11, 12].

Currently, observing bike helmet usage is primarily a manual task. While the reliability of this method is inherent in its simplicity, human factors can introduce error that skews the results of this approach. For example, time frame limitations on when human counters can be deployed may result in unfair representation of cyclist helmet usage. Collecting enough data on helmet usage to adequately model helmet use further requires longer observation periods, which may not be possible with manual counting alone. As a result, the human cost of such a survey can quickly become overbearing, and a consequence is that data collection of current surveys is limited.

Computer vision offers a solution to these challenges by automating the observation and analysis process. Unlike manual counting, computer vision systems can operate continuously and can be scaled efficiently to larger studies. Additionally, popular benchmark datasets [13, 14] include bicycles and helmets as detection classes, providing a good precedent for detecting these objects.

A challenge in applying object detection models to detecting cyclists and bike helmets is the presence of objects bearing a heavy resemblance. Vehicles such as mopeds, scooters, and other modes of transport that share traffic space with cyclists can often appear similar enough that they are counted as being cyclists. For an automated

system aiming to measure bike helmet usage, this results in false positives and inaccurate data being registered.

In our study, we aim to address this issue by introducing a loss rank mining approach to training object detection models. To our knowledge, little research has been done to investigate the applicability of computer vision in the context of bicycle helmet safety [15, 16, 17]. Thus, we further introduce a holistic method for counting with the use of trackers, presenting the following as our contributions:

- Using loss rank mining as a method of suppressing false positives, resulting in more accurate detection and tracking.
- Using a mix of the default YOLOv8 loss and loss rank mining during training to improve false positive suppression.
- The use of trackers and spatial information of detected objects to facilitate actual counting of cyclists and helmets in videos.

As we are collecting videos in a public setting, we blur faces in images presented in our work to protect privacy. In addition, to further ensure the integrity of cyclist privacy, the datasets used in this study are kept proprietary.

We begin with an overview of studies serving as the foundations of our research (Sec. 1.1). In the further chapters, we discuss the specific methodology we use in our approach (Sec. 2), experiments and their results (Sec. 3), and finally conclude our study with a discussion of our results and possible future work (Sec. 4).

1.1 Related Work

1.1.1 Cyclist detection & safety

A few studies exploring computer vision applied to cyclist safety are available in the literature. Authors of [18, 19, 20, 21, 22] apply various approaches to detect cyclists as a means of measuring their safety in spaces shared with vehicles. Contrastively, research by [23, 24] approaches cyclist safety measurement by probabilistic analysis of cyclist behaviour to determine whether cyclists may be in danger or not. While analysis strictly limited to cyclist helmet usage is scarce, authors of [17] have laid groundwork for analysis of this specific metric. Additionally, the work of [15, 25, 16] adds to the literature of Deep-Learning-based analysis measuring helmet usage in relevant context.

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In the same effort of improving cyclist safety, public datasets have been made available for benchmarking cyclist detection capabilities [26, 23, 27, 21]. These datasets introduce a diverse image set of cyclists in different orientations, different bicycles, and different helmets, providing a strong foundation of external data that can be used during training.

1.1.2 Object detection

Object detection models are classed into single-shot detectors [28] and two-shot detectors [29]. Between these approaches to object detection, single-shot detectors have shown to be more efficient in inference, capable of performing real-time detection on video recordings.

Models based on the YOLO architecture [30, 31] have been especially prominent as state-of-the-art single-shot object detection models, with YOLOv8 [32, 33] becoming the go-to object detection model for several applications, and the recently published YOLOv9 [34] and YOLOv10 [35] already demonstrating further performance improvements. Other advancements in object detection models have been made by RT-DETR [36], which builds upon work that integrates transformers into computer vision and object detection [37, 38].

While YOLO-based models have been built with a heavy emphasis on being generalisable, DETR-based models have shown to be less performant when applied to domains unseen during training [34]. This comes with implications for our research, as it can result in a significant number of false positives or false negatives. As this research focuses on false positives, we continue our discussion on false positive suppression methods proposed by other studies.

1.1.3 False positive suppression

Various approaches of false positive suppression have been developed for object detection. A common approach to suppressing false positives is done by analysing detections separately to determine which predictions are false positives, such as the work of [39, 40, 41]. In their study, the authors of [42] introduce a method of finding such false positives. The approach proposed by the above study tracks detections across frames. And by finding detections that have a short life-span across frames, i.e. detections that ‘flicker’, it is then possible to determine which detections are false positives and true positives.

Another approach is to change how the model learns to make is less prone to false positives in its predictions, such as research done by [43, 20, 44, 45]. A method that is particularly relevant for our study is the approach of [46]. The authors of this study introduce a new way of calculating the loss of YOLOv5 [47], so that it focuses on learning from examples that it performed worse on (measured by a loss function.) By altering the loss function to prioritise learning from examples where the model performed poorly, model predictions are expected to exhibit enhanced performance in challenging scenarios. Rather than focusing specifically on false positives, this approach instead encourages the model to focus more on instances where it struggles to make correct predictions, potentially leading to improved accuracy and robustness.

1.1.4 Object tracking and counting

Fundamentally, modern tracking algorithms work by assigning unique identifiers to objects detected in a frame, which can then be used to determine which objects are also present in the next frame. One approach that recent trackers are based on is the proposed method of DeepSORT [48]. Since its proposal, significant advancements in tracking algorithms have been developed. Among the notable ones are StrongSort [49], which improved the DeepSORT approach directly, ByteTrack [50], and BoT-SORT [51], each of which has demonstrated high performance on the MOT17 [52] and MOT20 [53] challenges.

With the use of trackers, it also becomes possible to count objects across image frames. A common approach is to use the number of unique identifiers a tracker makes, and using that as the number of objects counted. The authors of [54] evaluated two different methods



Fig. 1: Example image with annotations for cyclists and bike helmets.

Distribution	Dataset Name	Images	Cyclists	Helmets
Train	Acq 2022	2519	3609	1605
	CIMAT [23]	2259	3345	2599
	External – 1	550	847	379
Validate	Westerplantage	9873	2229	62
Test	Voorstreek	9257	2040	110

Table 1: General description of locally acquired and external datasets. Validation and Test sets include negative frames to imitate intended deployment conditions when loading data sequentially.

of counting using trackers. By further introducing a ‘time-alive’ consideration for counting objects, similar to the approach of [42], tracked objects that don’t persist for multiple frames are discarded.

The other approach relies on a Region of Interest (ROI) line that divides each frame. This approach incorporates spatial information of each frame by counting only objects that have been present on both sides of the line. In doing so, the tracker has further possibility of correcting erroneous detections, and stationary objects can be ignored.

2 MATERIALS AND METHODS

2.1 Dataset

The model was trained on a diverse dataset from videos captured locally, as well as two external datasets. Each cyclist and bike helmet in these images are individually annotated with a bounding box, as shown in Fig. 1. Table 1 describes the datasets collected both from our acquisitions and external dataset.

2.1.1 Captured datasets

To include locally acquired data in our experiments, recordings were done across different locations captured in the province of Friesland. Training data was collected around the centre of Leeuwarden from points indicated in Fig 2. At each of these locations, a GoPro camera was situated parallel to paths that regularly see cyclist traffic.

In addition to the datasets used for training and testing, more videos were captured in other localities of Friesland to create a counting set (2). These recordings are used to compare the counting capabilities of our approach to the actual counts of each video, making it possible to measure counting performance. Per-frame annotations were therefore not required for these recordings, only counts of cyclists and helmets in each video

2.1.2 External datasets

To improve the robustness of our models, more training data was collected from external sources. We made use of the CIMAT [23] dataset, which was originally created to train models that determine

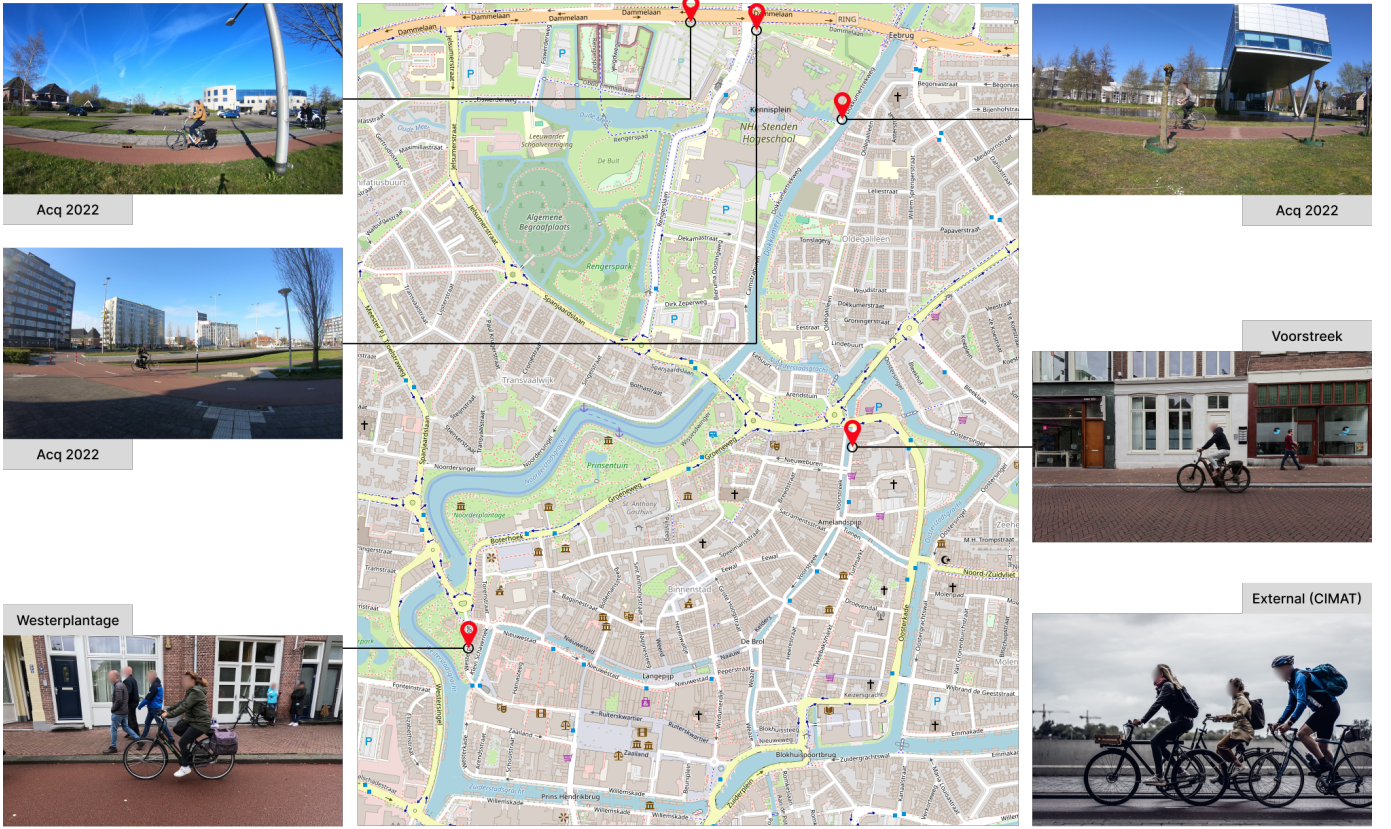


Fig. 2: Sample images from datasets used in training, together with locations for locally captured recordings (© OpenStreetMap contributors).

Location Data	Cyclists	Helmets
Dokkum	50	0
Gytsjerk	40	1
Herbaijum	28	2
Sneek	203	16
Bolsward	123	18
Franeke	52	3
Heerenveen	110	0

Table 2: Total counts of cyclists and helmets in the additionally recorded videos.

cyclist orientation. To keep these images consistent with our own data, we leveraged the known cyclist direction in this dataset to select only cyclists facing left or right. The original bounding boxes of this dataset were kept for cyclists and renamed to be consistent with our data, while additional bounding boxes for helmets were then added.

In addition to the CIMAT dataset, more images were scraped from external sources. This dataset also comprised images of cyclists, but the orientation of cyclists was not considered. The majority of images in this dataset were of different helmets a cyclist may wear to add more diversity to what the models see during training, allowing them to be more general in their detections.

2.2 Object detection model

Since architectures based on the YOLO architectures have shown to be reliable in relevant work [55, 56, 57, 33, 17, 16], we use YOLOv8 in our experiments. The small variant is used with weights pretrained on the Microsoft COCO (Common Objects in Context) dataset [13]. While the COCO dataset contains the ‘Bicycle’ class in its labels, whether bicycles are in use is not considered, nor is the ‘Helmet’ class included in the dataset. Transfer learning was then used to steer

model detections toward bicycles that are actively in use under the ‘Bicycle’ class, and to train detections of the ‘Helmet’ class.

During the training process, mixed positive tiling was employed to augment the dataset and enhance the model’s exposure to data variability. This technique involved generating positive tiles from images, with negative tiles randomly mixed in at a 1:5 positive-to-negative ratio (0.2 negative tiles per positive tile). Two tiles of 896×896 pixels were generated per image. Additionally, random horizontal flipping was applied to images with a 0.2 probability during training to further diversify the training data.

2.3 False positive suppression

Since other studies have found loss rank mining to be effective in teaching YOLO-based models to learn from harder examples [43, 58], we used a similar approach in our study. In particular, we implemented methodology similar to that used by [46]. Our initial tests with the same model used by the authors showed promise in this approach to reducing false positive predictions.

Having established this as a possible method of suppressing false positives, we implemented a similar loss rank mining approach to YOLOv8. The default loss calculation is composed of a bounding box loss $\mathcal{L}_{\text{box}}$, a class loss $\mathcal{L}_{\text{cls}}$ and a Distributed Focal Loss (DFL) [59] $\mathcal{L}_{\text{dfl}}$, calculated as shown in (1).

$$\mathcal{L}(\theta) = \frac{\lambda_{\text{box}}}{N_{\text{pos}}} \mathcal{L}_{\text{box}}(\theta) + \frac{\lambda_{\text{cls}}}{N_{\text{pos}}} \mathcal{L}_{\text{cls}}(\theta) + \frac{\lambda_{\text{dfl}}}{N_{\text{pos}}} \mathcal{L}_{\text{dfl}}(\theta) + \phi \|\theta\|_2^2 \quad (1)$$

Here, the individual loss weights are incorporated with a regularisation term $\|\theta\|_2^2$ and weight decay ϕ .

Of these losses, DFL loss $\mathcal{L}_{\text{dfl}}$ teaches the model to learn location values of target bounding boxes, making it optimal for loss rank mining and learning hard examples. We focus specifically on the DFL loss in our study, which is calculated by YOLOv8 as (2).

$$\mathcal{L}_{dfl}(\theta) = \sum_{x,y} \mathbb{1}_{c_{x,y}^*} \left[- \left(q_{(x,y)+1} - q_{x,y} \right) \log(\hat{q}_{x,y}) \right] \quad (2)$$

The terms of (2) are defined as follows:

- $q_{x,y} = \text{IoU}_{x,y} = \frac{\beta_{x,y} \cap \hat{\beta}_{x,y}}{\beta_{x,y} \cup \hat{\beta}_{x,y}}$: The Intersection over Union (IoU) between the predicted box $\hat{\beta}_{x,y}$ and the ground truth box $\beta_{x,y}$ at position (x,y) .
- $\mathbb{1}_{c_{x,y}^*}$: An indicator function that is 1 if $(x,y) \in c_{x,y}^*$ and 0 otherwise, where $c_{x,y}^*$ represents the set of closest predicted box IoUs (left and right).
- $q_{(x,y)+1}$ and $q_{(x,y)-1}$: The nearest predicted box IoUs to the left and right of $q_{x,y}$.
- $\hat{q}_{x,y}$: The predicted IoU at position (x,y) .

We change the original DFL loss calculation (2) so that it takes only the highest $n\%$ of values during summation, as shown below.

The output loss of a single sample is first calculated as (3).

$$\delta_{x,y} = \mathbb{1}_{c_{x,y}^*} \left[- \left(q_{(x,y)+1} - q_{x,y} \right) \log(\hat{q}_{x,y}) \right] \quad (3)$$

We then define conditions that limit selected loss values to the top $n\%$ of calculated values (4).

$$\mathbb{1}_{x,y}^{n\%} = \begin{cases} 1 & \text{if } \delta_{x,y} \in \text{top } n\% \text{ of all } \delta_{x,y} \text{ values} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

The loss values are then collected and summed to form the final DFL loss value, with an $n\%$ of 35% (5).

$$\mathcal{L}_{dfl}(\theta) = \sum_{x,y} \mathbb{1}_{x,y}^{n\%} \cdot \delta_{x,y} \quad (5)$$

In addition to introducing loss rank mining, to ensure that the model learns easier representations before focusing on hard examples, we introduce mixed training that uses both the default Distributed Focal Loss and the mined loss. In this approach, the model is trained for the first half using the default loss, and the mined loss is used afterward.

2.4 Counting algorithm

Counting cyclists and helmets was done by employing a tracker together with an object detection model. This method uses appearance features of detected objects in addition to temporal information across frames to determine which objects to track. By assigning a unique identifier to objects that are present in multiple frames, each object further becomes a unique object spanning multiple frames. We utilise the BoT-SORT [51] tracker for counting, as preliminary investigation showed that it outperformed Strong-SORT [49] and ByteTrack [50].

For the actual counting of objects, we found unique identifiers to not be sufficient on their own, as objects can be assigned different identifiers when their features change, causing double counts as a result. To counter-act this, we introduce a counting approach similar to [54]. A region-of-interest (ROI) line is drawn vertically to divide each frame, and the position of each object relative to this line is assessed, as shown in Fig. 3. By counting only objects that have been present on both the right and left side of this divider, a more reliable approach to counting the objects is achieved. To further improve the robustness of the counting algorithm, we implement a time-alive approach to limit counted objects to those present for at least five frames.

2.5 Metrics

In this study, metrics that evaluate the object detection, tracking, and counting performance are employed. We discuss these metrics in this section.

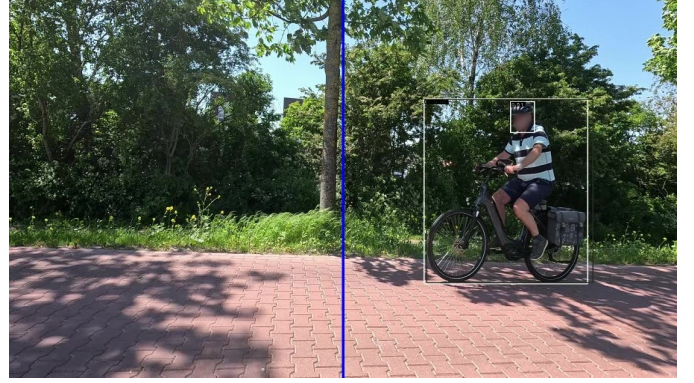


Fig. 3: Example of the ROI line dividing the image.

2.5.1 Object detection metrics

The standard metric used in evaluating performance of object detection algorithms is the *mean Average Precision* or *mAP*. The *mAP* is defined using both the Precision and Recall metrics, where Precision refers to the proportion of correctly predicted positive samples out of all samples that the model predicted as positive. Precision is calculated

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP is the number of True Positives, and FP is the number of False Positives. We place further emphasis on the False Positive statistic, in our study, as that is one of the focal measurements we look to minimise in our approach. Recall, on the other hand, quantifies the percentage of true positive instances the model correctly detects out of all actual positive instances in the datasets. Formally, this is calculated

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN is the number of False Positive predictions made by the model. For object detection models, a higher Recall and Precision indicate better overall performance. In practice, however, there is often a trade-off between the two, as increasing one often leads to a decrease in the other.

Instead, the F1 Score measures a balance between the precision and recall, measured as

$$F1 \text{ Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

In addition to F1 Score, *mAP* is often used in object detection to plot the precision at different recall levels. When calculating *mAP*, the Average Precision (AP) is computed as the area under the curve of the precision-recall curve. By taking the mean AP across all classes, *mAP* is then calculated

$$mAP = \frac{1}{C} \sum_{c=1}^C AP_c, \quad \text{where } AP = \int_0^1 f_{pr}(r) dr$$

where C is the number of classes, and f_{pr} is given by the precision-recall curve at model confidence ranging from 0 to 1.

2.5.2 Tracking and counting metrics

To measure the performance of object tracking models, two standard metrics are used [52]. Multiple Object Accuracy (MOTA) is a metric that quantifies the error ratio of the tracker, encompassing misses, false positives and mismatches relative to the total ground truth objects. It is calculated as

$$MOTA = 1 - \frac{\sum_t (FN_t + FP_t + IDSW_t)}{\sum_t (GT_t)},$$

Model	F1 Score	mAP
Default	0.82 (± 0.02)	0.35 (± 0.03)
Mixed	0.85 (± 0.02)	0.35 (± 0.02)
LRM	0.77 (± 0.07)	0.34 (± 0.02)

Table 3: Mean and standard deviation of F1 Score and mAP of the default, mixed, and LRM models over five trials.

where t refers to the frame index relative to the total number of frames, and GT refers to ground truth objects in the frame at time t . $IDSW$ in this case, refers to identity switches that happen when the tracker mistakenly assigns different identities to the same object across consecutive frames.

While MOTA provides a comprehensive evaluation of tracking performance by considering misses, false positives, and identity-switches, Multiple Object Precision (MOTP) focuses on the accuracy of object localisation. MOTP is calculated as

$$MOTP = \frac{\sum_t d_{t,i}}{\sum_t c_t},$$

where c_t represents the number of matches in from t and $d_{t,i}$ represents the bounding box overlap of target i with its assigned ground truth object.

Finally, counting accuracy is evaluated by calculating the ratio of correctly counted instances against the total instances:

$$Count\ Accuracy = \frac{TP}{TP + FP + FN}.$$

Because this accuracy is in the context of the object detection task, True Negatives (TN) have been omitted from the accuracy equation.

3 EXPERIMENTS & RESULTS

3.1 Object Detection

The model was trained for 100 epochs on the previously described data, and tested to measure object detection performance. In our experiments, we test three models. The first is the default YOLOv8 model trained using the original loss calculation, a mixed model using both the original loss and loss rank mining loss, and a model that purely uses loss rank mining (default, mixed, LRM).

For the first experiment, five trials were done to test object detection performance on the test set described in Tab. 1. Overall, results on the F1 Score and mAP (Tab. 3) showed similar performance for each model, with the mixed model performing slightly better than the other two. On the other hand, results on false positive counts (Fig. 4) showed that mixed training demonstrated a significantly lower mean false positive count for cyclists, while reducing false positive counts for helmets to a much lesser extent. Notably, the LRM model performed worse on average, with significantly more deviation than the other two models.

Analysis of the predictions made by the default and mixed models 5 showed that the use of loss rank mining had an impact on false positive detections made that would normally confuse the model.

Based on the above results, we move forward to tracking and counting experiments by comparing the best models of the default and mixed approach.

3.2 Tracking

To test model tracking, we compare the best default and mixed models from the previous experiment, testing performance using MOTA and MOTP. Tracking was tested using the test set described in Tab. 1, similar to the previous experiment. Because this experiment tests tracking capabilities, data was loaded sequentially to keep temporal information intact. As mentioned in Sec. 2.4, a pretrained BoT-SORT tracker is used to track detections across frames.

Results of this experiment (Tab. 4) showed that both models measured a similar MOTP, indicating similar levels of localisation for

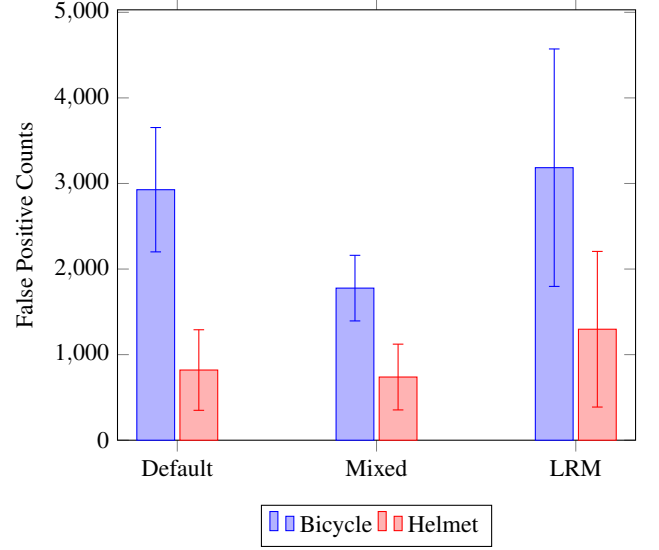


Fig. 4: Mean false positive counts for the default, mixed, and LRM models over five trials.

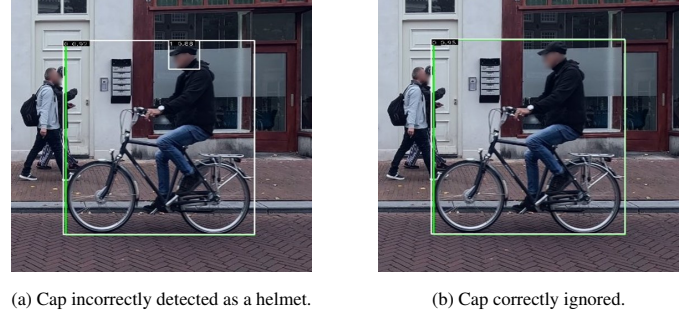


Fig. 5: Comparison between cap incorrectly detected as a helmet by the default mode (5a), and the same cap correctly being ignored by the mixed model (5b).

Model	Class	MOTA	MOTP
Default	Cyclist	0.58	0.10
	Helmet	0.18	0.30
Mixed	Cyclist	0.66	0.10
	Helmet	0.34	0.30

Table 4: MOTA and MOTP results of the default and mixed models across different classes.

Location	Class	TP	FP	FN	Accuracy
Dokkum	Cyclists	50	8	3	81.97%
	Helmets	0	0	0	-
Gytsjerk	Cyclists	37	2	2	90.24%
	Helmets	0	1	1	0.0%
Herbaijum	Cyclists	26	1	2	89.66%
	Helmets	0	1	2	0.0%
Sneek	Cyclists	188	2	11	93.53%
	Helmets	6	1	10	35.29%
Bolsward	Cyclists	111	4	12	87.40%
	Helmets	15	16	4	42.86%
Franeker	Cyclists	44	0	8	84.62%
	Helmets	0	0	3	0.0%
Heerenveen	Cyclists	105	0	5	98.46%
	Helmets	0	1	0	0.0%
Summary	Cyclists	561	17	43	90.34%
	Helmets	21	20	20	34.43%

Table 5: Results of the full counting process applied to locally captured videos. Accuracy in the summary is calculated based on its respective summated TP, FP and FN values, rather than averaged based on the location accuracies.

tracked objects. However, the mixed model showed significantly higher MOTA for both classes, indicating that the tracked objects were much closer to the annotated objects and better tracking capability. For the Cyclist class, tracking performance increased by 8%. For the Helmet, class, however, performance increased significantly more compared to the default model, with a 16% MOTA improvement.

3.3 Counting

To measure counting performance, the mixed model used in the previous experiment was selected and tested on the extra videos captured, as detailed in Tab. 2. Each counted object was then assessed individually to determine the accuracy of the model when counting.

The resulting counts produced by the model (Tab. 5) show a relatively low deviation from the actual counts. Further analysis, however, showed that predictions were sometimes closer to the actual counts by coincidence. For example, the counts in Gytsjerk appeared to be exactly the same as the actual counts, but results showed that the model falsely predicted two other vehicles as bicycles, and missed two actual bicycles. This is shown in Tab. 5 where the number of false positives and false negatives are even, resulting in a correct count output. The same was the case for the helmet detected.

In addition, an important detail is the low presence of helmets apart from the Bolsward and Franeker locations, making accuracy calculations less reliable for individual locations. In the two locations with more helmet presence, as well as the overall count results, accuracy could be calculated much more reliably.

4 DISCUSSION, CONCLUSION & FUTURE WORK

4.1 Discussion

Throughout our experiments, we noticed the presence of several factors that affected results.

While measuring tracking performance, certain scenarios affected how the model made its predictions. For example, some cases where cyclists rode too close to the camera or rode past too quickly, were missed by the model due to them not being present in enough frames.

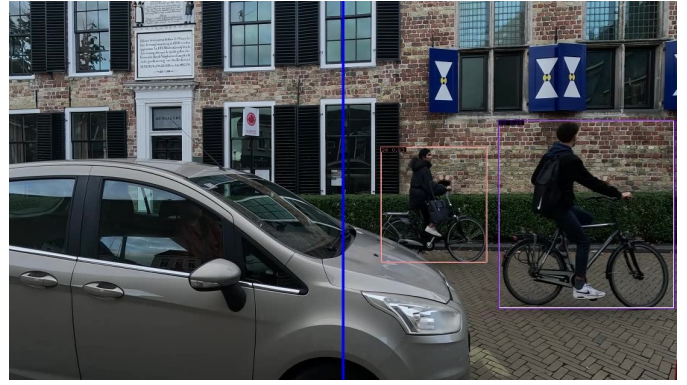


Fig. 6: Example edge case in the Franeker location, where a part of the frame is blocked, interfering with the counting process.

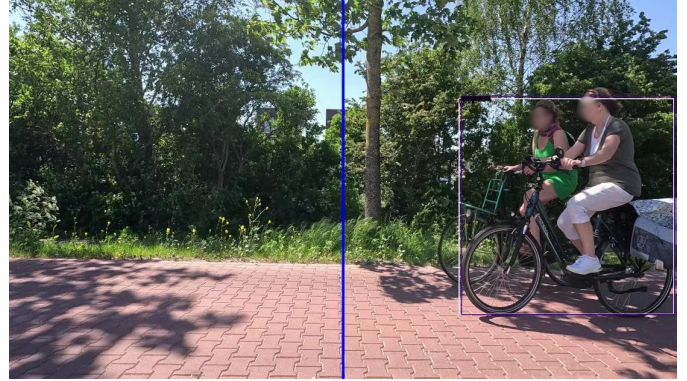


Fig. 7: Example case where cyclists are side-by-side, resulting in missed detections by the model.

While this was the case for cameras situated closer to cycle paths, the opposite was observed when the camera was situated much further away.

Additionally, cases where objects linger in front of the camera (Fig. 6) and where cyclists ride side by side (Fig. 7) resulted in the model missing predictions. This resulted in several false negatives, particularly in busier bike paths like the ones in Sneek and Bolsward from our counting test set. While technical solutions to this are limited, the setup of the camera can be adjusted to account for these issues. For example, the camera can be situated in a way that larger vehicles such as cars and trucks do not have the possibility of obscuring view. A higher position can be further used for the camera to achieve an angle that captures cyclists riding side by side without them obscuring each other.

Another trend we observed was that cyclists tended to look directly into the camera, particularly if the camera stood out significantly during recordings. When this occurred, the model often predicted the cyclist to be wearing a helmet, such as in the Bolsward location indicated in 5, resulting in heavily skewed helmet counts. This contrasted with locations where the camera was more concealed, such as the Dokkum location, where the model correctly ignored all head-gears.

4.2 Conclusion

This study aimed to create a holistic approach to counting cyclists and helmets for automating cyclist and helmet counting. A camera setup parallel to a cycle path is used to acquire recordings of cyclists and measure helmet use in situations with limited potential interference. We further aimed to use loss tank mining to reduce false positive detections caused by other vehicles.

Our experiments showed that the proposed mixed training

approach is capable of teaching models to reduce false positive predictions, as well as a slight improvement in overall performance. This improvement complements the tracking capabilities when combined with the BoT-SORT tracker and significantly improves tracker accuracy, measured by using MOTA, significantly improving tracking of both cyclists and helmets across frames. Further, by introducing an ROI line to track relative object movement, false positive detections can be corrected before they are counted.

Overall, this approach represents a promising step forward in automating the task of measuring cyclist helmet usage.

While further refinement can be done, particularly in improving missed detections, the significant increase in performance provided by our approach highlights its potential for cyclist and helmet detection.

4.3 Future Work

In our study, we introduced mixed loss training as a way of reducing false positives, which proved to be effective in doing so. However, room to explore other approaches to mixed loss training is still abundant. The approach used in our experiments applied hard example mining halfway through training, but different approaches may ensure the model learns even better and more balanced representations of easy and hard examples.

We also note that different backgrounds resulted in more erratic predictions by the model. While it is not clear how the background influences model predictions, it is worth exploring in future work to ensure accurate counts.

Further, in our work, we select images from the CIMAT [23] dataset where cyclists are facing either left or right. However, we noticed that our approach to data collection often results in cyclists looking into the camera occasionally. Incorporating other images from this dataset, where cyclists are facing into or away from the camera, may allow the model to better distinguish head-gear and helmets.

Another area we see room for future work is the use of spatial information to facilitate counting. Our study used an ROI line to divide images and leverage the movement of cyclists, but as discussed, this approach fails to address certain cases where this information cannot be used effectively.

In our experiments, measuring counting (Sec. 3.3), we raised the concern of counts sometimes appearing to be correct by coincidence, introduced by the tendency of models to predict false positive or false negative objects. While technical solutions to this are limited, we propose measuring the error rate as a metric for future projects aiming to utilise counting. By doing so, performance in practice can be measured more accurately.

Finally, while we test our approach to loss rank mining using YOLOv8, the recently released YOLOv9 [34] uses the same loss calculation as YOLOv8, making it an optimal model for further testing with loss rank mining and our approach with mixed training.

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